

# Stock Earnings Agent

## Project Documentation

**Version:** 1.0 **Date:** September 2026 **Status:** Production ready

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## 1. Overview

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Stock Earnings Agent is an equity research assistant that answers questions about public companies by combining three sources of truth: live market data, official regulatory filings, and a searchable library of earnings documents. It reasons over these sources with a large language model and presents results through two interfaces: an interactive web application and a Model Context Protocol (MCP) server that any MCP compatible client can call.

The system is built as a set of clearly separated layers so that the same tested business logic serves both the web application and the MCP server, with no duplicated code.

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## 2. Problem Statement

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Analysing a company's earnings is slow and fragmented. An analyst typically has to:

1. Pull quantitative metrics (revenue, margins, valuation ratios) from one source.
2. Read qualitative commentary (management discussion, guidance, risk factors) from official filings in another place.
3. Manually reconcile the numbers with the narrative to form a view.

These steps live in different tools, use different formats, and require repeated manual effort for every company and every quarter. There is no single interface that retrieves the numbers, retrieves the relevant text, and reasons over both together.

**Goal:** provide one system that gathers the quantitative and qualitative evidence for a company, retrieves the most relevant passages from earnings documents, and produces a grounded, cited analysis, available both as a web application and as callable tools for an AI assistant.

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### 3. Objectives

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| # | Objective                     | Outcome                                                               |
|---|-------------------------------|-----------------------------------------------------------------------|
| 1 | Aggregate live financial data | Fundamentals, prices, ratios and analyst views from a single tool set |
| 2 | Access official filings       | On demand retrieval of 10-Q, 10-K and 8-K documents from SEC EDGAR    |
| 3 | Search earnings documents     | Retrieval augmented generation over ingested PDFs and transcripts     |
| 4 | Reason over the evidence      | A tool calling agent that plans, retrieves, and answers               |
| 5 | Expose tools to AI clients    | An MCP server that reuses the same logic                              |
| 6 | Measure retrieval quality     | An evaluation harness with retrieval and generation metrics           |
| 7 | Ship to production standard   | Typed configuration, logging, tests, linting and containerisation     |

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### 4. Solution Overview

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The solution is organised into five layers plus two cross cutting concerns.

- **Data sources** provide raw inputs: market data, filings, and documents.
- **RAG pipeline** converts documents into a searchable vector index.
- **Shared tool layer** holds the business logic as plain functions.
- **Consumers** are the MCP server and the reasoning agent, which both call the shared tools.
- **Presentation** is the web application.

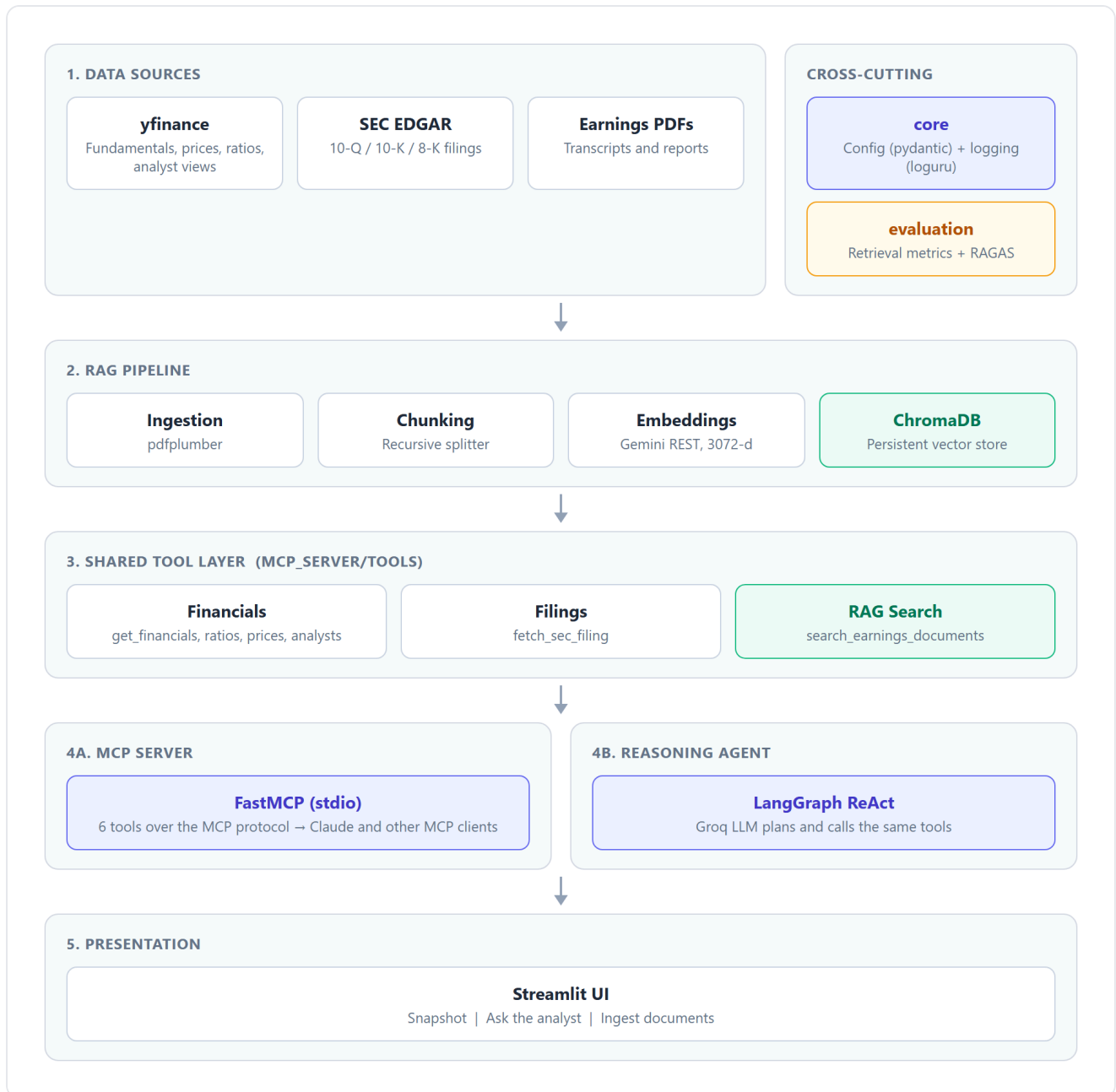
The two cross cutting concerns are **core** (configuration and logging) and **evaluation** (quality measurement).

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### 5. System Architecture

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The diagram below shows the layered design and how data flows from raw sources to the user. The shared tool layer is the key design decision: both the MCP server and the agent depend on it, so logic is written and tested once.



## Design principles

- **Single source of business logic.** Financial, filing and retrieval logic live in `mcp_server/tools/` as framework neutral functions. The MCP server and the agent are thin adapters over them.
- **Predictable failure.** Tool functions never raise for expected problems such as a bad ticker or a network error. They return a structured `{"error": ...}` value so callers can reason about it.
- **Configuration in one place.** All settings are validated once at startup through a typed settings object.
- **Logs to standard error.** The MCP protocol uses standard output, so all logs are written to standard error to keep the protocol stream clean.

## 6. Technology Stack

Every dependency is chosen for a specific role. Links point to the official documentation.

| Area            | Technology                                                         | Role                                          |
|-----------------|--------------------------------------------------------------------|-----------------------------------------------|
| Language        | <a href="#">Python 3.11+</a>                                       | Implementation language (validated on 3.13)   |
| LLM inference   | <a href="#">Groq</a>                                               | Fast hosted inference for the reasoning model |
| Agent framework | <a href="#">LangGraph</a>                                          | Stateful tool calling ReAct agent             |
| LLM integration | <a href="#">LangChain</a>                                          | Tool and model abstractions                   |
| Embeddings      | <a href="#">Google Gemini Embeddings</a>                           | 3072 dimensional text vectors via REST        |
| Vector store    | <a href="#">ChromaDB</a>                                           | Local persistent similarity search            |
| Tool protocol   | <a href="#">Model Context Protocol</a> via <a href="#">FastMCP</a> | Exposes tools to AI clients                   |
| Market data     | <a href="#">yfinance</a>                                           | Fundamentals, prices, ratios, analyst views   |
| Filings         | <a href="#">SEC EDGAR</a>                                          | Official 10-Q, 10-K and 8-K documents         |
| PDF parsing     | <a href="#">pdfplumber</a>                                         | Text extraction from earnings PDFs            |
| Web UI          | <a href="#">Streamlit</a>                                          | Interactive front end                         |
| Evaluation      | <a href="#">RAGAS</a>                                              | RAG generation quality metrics                |
| Configuration   | <a href="#">Pydantic Settings</a>                                  | Typed, validated settings                     |
| Logging         | <a href="#">Loguru</a>                                             | Structured application logging                |
| Resilience      | <a href="#">Tenacity</a>                                           | Retry with backoff on transient errors        |
| Testing         | <a href="#">pytest</a>                                             | Unit and integration tests                    |
| Linting         | <a href="#">Ruff</a>                                               | Static analysis and formatting                |

### Notable decisions

- Embeddings are generated by direct REST calls to the Gemini API rather than local transformer libraries. This avoids native library incompatibilities on Windows with Python 3.13 and removes a heavy dependency.
- The reasoning model is configurable. The default is a tool calling model available on the target Groq account. It can be changed with one environment variable.

## 7. Core Components

This section explains the major pieces of the codebase. Each subsection names the files involved and shows a short representative excerpt.

### 7.1 Configuration and logging ( `core/` )

All configuration is defined once as a typed object. Validation runs at startup, so a missing key or an invalid value fails immediately with a clear message rather than deep inside a request.

```
class Settings(BaseSettings):
    groq_api_key: str = Field(..., description="API key for Groq inference.")
    google_api_key: str = Field(..., description="API key for embeddings.")
    groq_model: str = Field("qwen/qwen3.8-27b") # must support tool calling
    embedding_model: str = Field("gemini-embedding-001")
    embedding_dimension: int = Field(3072, gt=0)
    chunk_size: int = Field(1000, gt=0)
    retrieval_top_k: int = Field(5, gt=0)
```

Logging is centralised in `core/logging.py` and writes to standard error, which keeps the MCP protocol stream on standard output uncorrupted.

### 7.2 RAG pipeline ( `rag_pipeline/` )

The pipeline turns documents into a searchable index and answers retrieval queries. It is split into focused modules:

- `ingestion.py` reads PDF, text and markdown files into documents with provenance metadata.
- `chunking.py` splits documents into overlapping windows using a recursive splitter.
- `embeddings.py` calls the Gemini API to produce vectors, with batching and retries.
- `vector_store.py` wraps ChromaDB for persistent similarity search.
- `pipeline.py` orchestrates the write path and the read path.

The embedding client retries transient failures automatically:

```
@retry(
    retry=retry_if_exception_type((requests.RequestException, EmbeddingError)),
    stop=stop_after_attempt(4),
    wait=wait_exponential(multiplier=1, min=1, max=10),
    reraise=True,
)
def _embed_one(self, text: str, task_type: str) -> list[float]:
    ...
```



The system prompt instructs the model to fetch data with tools before making any claim, to cite retrieved passages, and to close with a short labelled summary.

## 7.5 MCP server ( `mcp_server/server.py` )

The server is a thin adapter that registers each shared function as an MCP tool. Business logic stays in the tool modules.

```
@mcp.tool()
def get_financials(ticker: str) -> dict[str, Any]:
    """Fetch key fundamentals for a ticker."""
    return tools.get_financials(ticker)
```

## 7.6 Web application ( `app/main.py` )

The Streamlit application has three tabs: a quantitative snapshot, a conversational analyst backed by the agent, and a document ingestion panel that grows the searchable library.

## 7.7 Evaluation ( `evaluation/` )

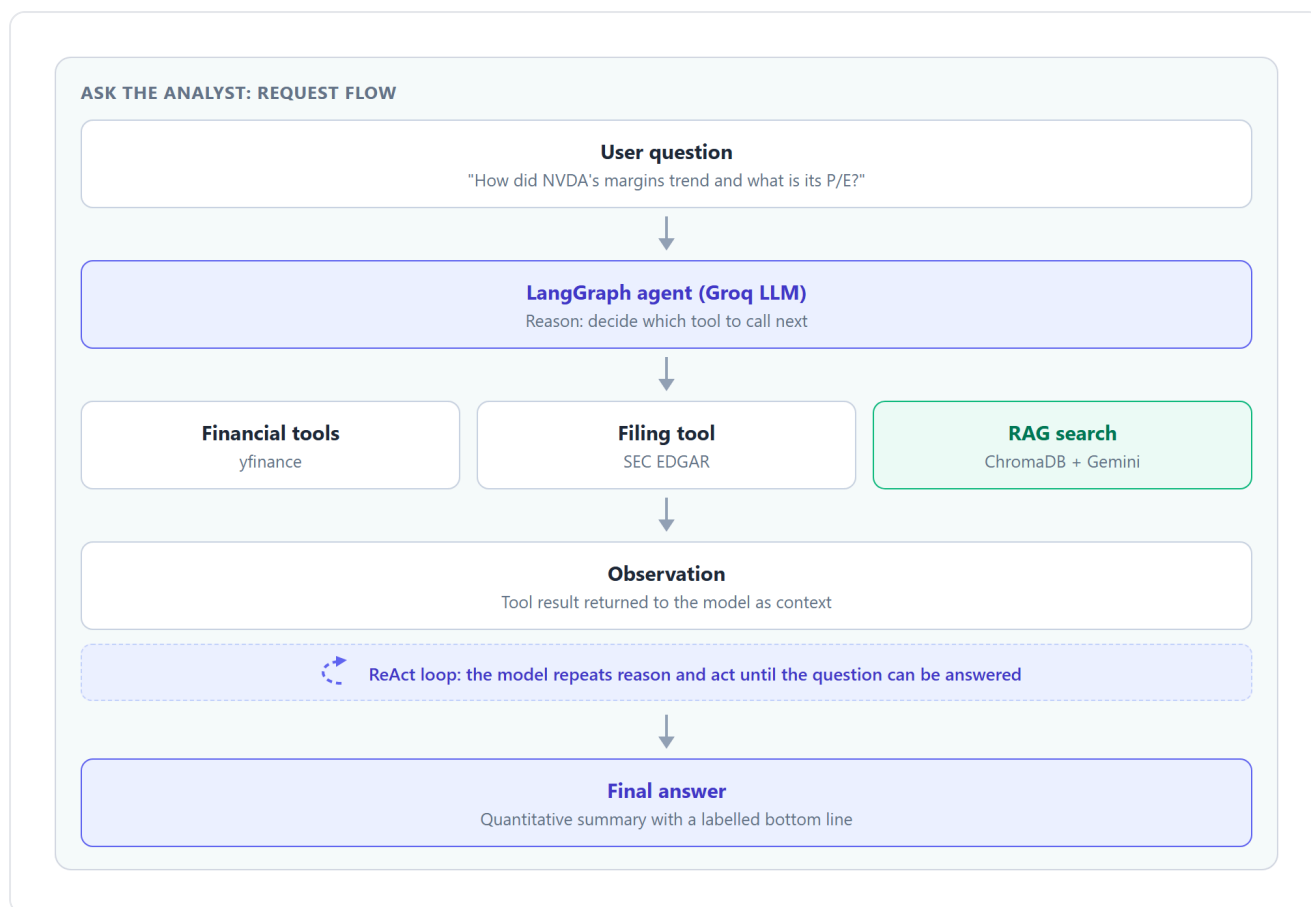
Two complementary evaluators are provided. A deterministic retrieval evaluator computes hit rate and mean reciprocal rank without using any model tokens. A RAGAS based harness adds generation metrics such as faithfulness and answer relevancy when an inference budget is available.

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# 8. Request Flow: Ask the Analyst

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The diagram below shows what happens when a user asks a question. The model decides which tool to call, the tool runs, the result is fed back as context, and the loop repeats until the model can produce a final grounded answer.



## 9. Application Walkthrough

### 9.1 Quantitative snapshot

The snapshot tab retrieves fundamentals, price performance, valuation ratios and the analyst consensus for a ticker, then renders them as an interactive, visual dashboard: headline metric cards, speedometer gauges and six finance charts, with the exact figures preserved in a collapsible table.



# Stock Earnings Agent

Snapshot Ask the analyst Ingest documents

## Quantitative snapshot

Search by company name or ticker symbol.

The full result for a sample ticker is shown below.



# Stock Earnings Agent

Snapshot Ask the analyst Ingest documents

## Quantitative snapshot

Search by company name or ticker symbol.

Company

Apple Inc. (AAPL)

Price period

1y

Fetch snapshot

## Apple Inc.

Technology · Consumer Electronics

Market cap

4.87T

P/E (TTM)

38.25

Profit margin

27.6%

Revenue growth

16.4%

Price

\$333.53

1y high

\$341.07

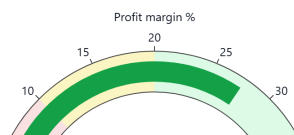
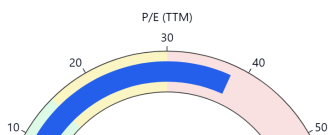
1y low

\$244.37

Avg volume

48.08M

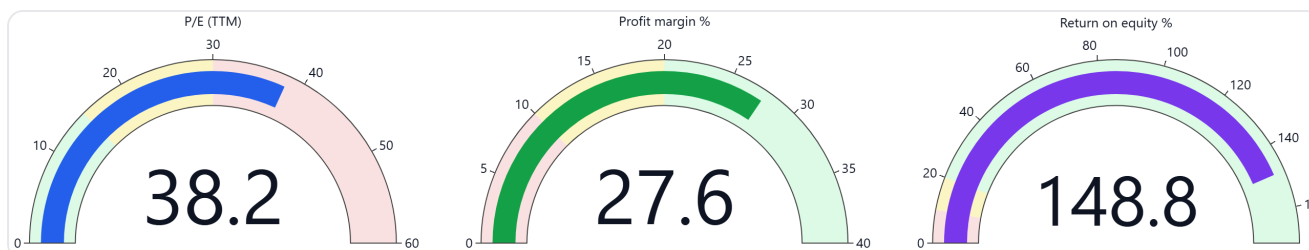
↑ 30.19%



## 9.2 Visual analytics

The snapshot is visual first, following the conventions professional equity platforms use: a clean layout, consistent colour coding (green favourable, red unfavourable, blue accent, grey context), hover interactivity and generous whitespace. The charts are built with Plotly and render directly in the web app. Each one answers a specific analytical question.

### Headline KPI gauges



**What it shows.** The three numbers that frame a first look — P/E (TTM), profit margin and return on equity — each on a speedometer with coloured comfort bands. The scale auto-expands for outliers (for example a return on equity near 150%) so the needle never saturates at the end of the dial.

**Where it is used.** An executive-style read: is the stock cheap or expensive, and is the business profitable and efficient, in one glance.

### Price and volume

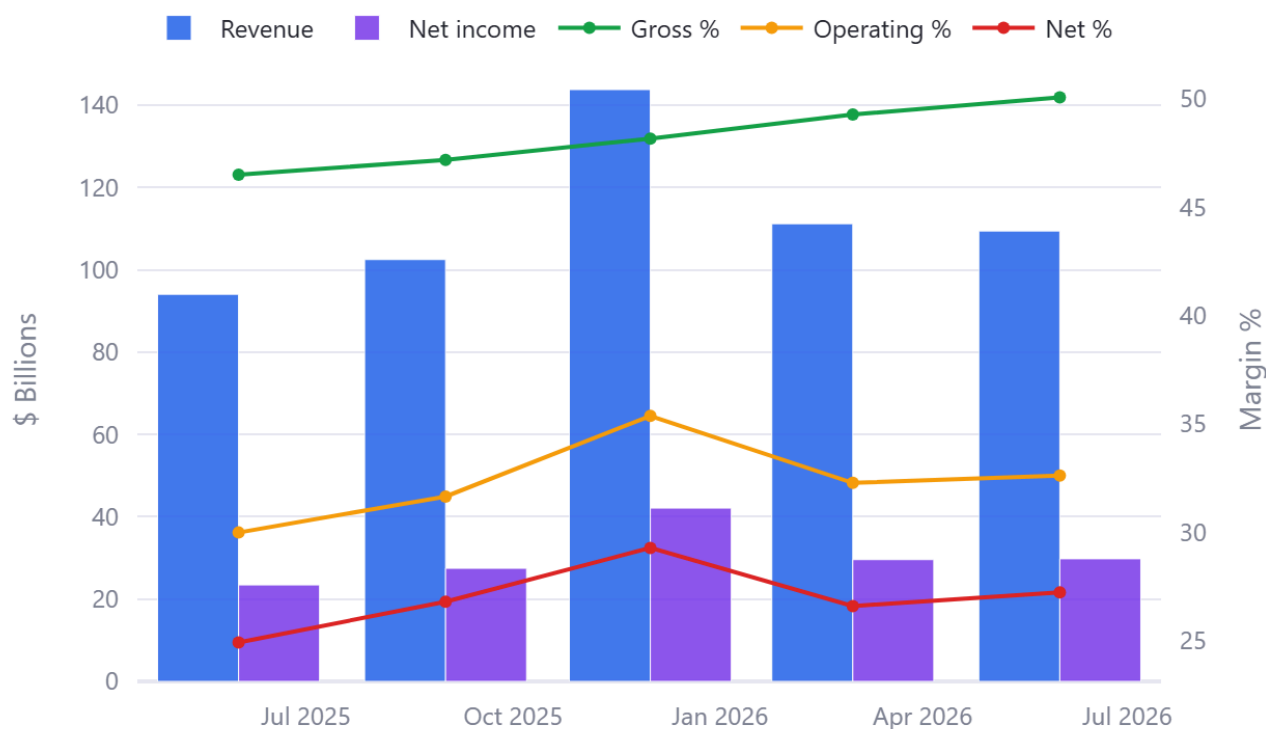


**What it shows.** A candlestick of daily open, high, low and close over the selected period, with a volume sub-panel coloured by up and down days. The title carries the period return. Candlesticks convey trend, volatility and the balance of buyers and sellers that a single closing-price line hides.

**Where it is used.** Reading momentum and spotting the days where price moved on unusually heavy volume.

### Revenue, net income and margins

## Revenue, net income & margins

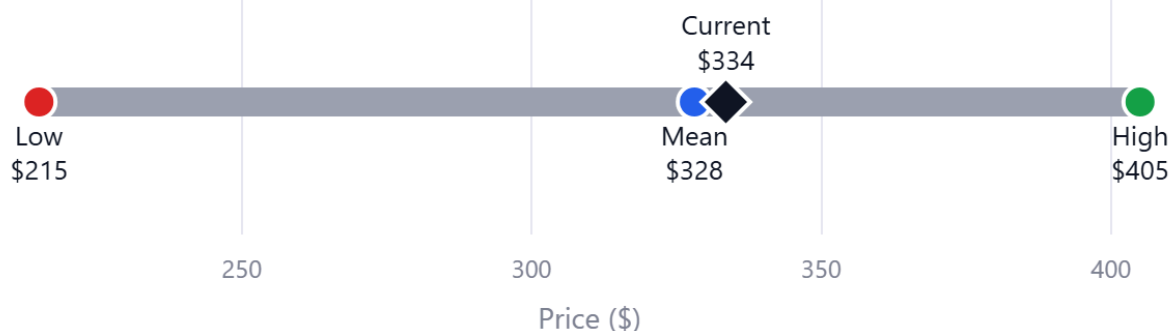


**What it shows.** Quarterly revenue and net income as grouped bars, with gross, operating and net margin lines on a second axis. Plotting the absolute dollars beside the margin percentages reveals whether growth is actually translating into profit — widely regarded as the single most useful view in fundamental analysis.

**Where it is used.** The core fundamental read: is the company scaling efficiently, and are margins expanding or compressing over time.

## Analyst price targets

### Analyst price targets (-1.6% to mean)

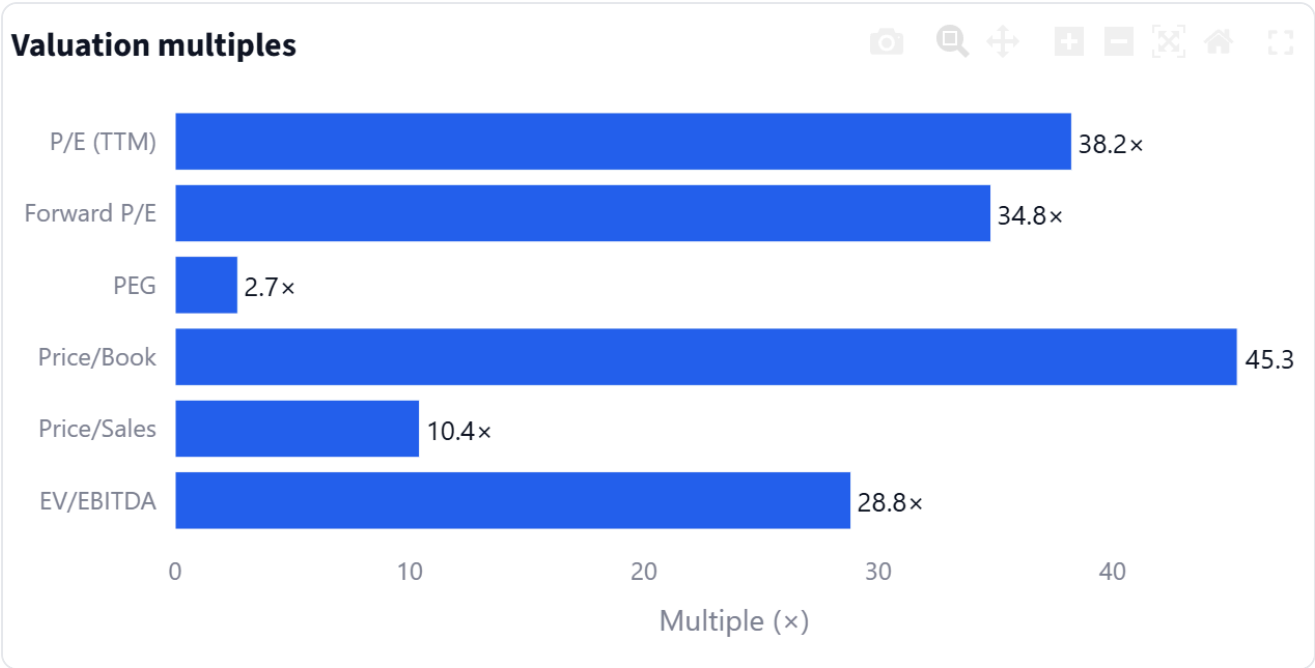


**What it shows.** The low, mean and high analyst price targets plotted as a range against the current price, annotated with the implied upside or downside to the mean. The spread between low and high

captures how much analysts disagree.

**Where it is used.** A one-glance sense of the market's expected range and where today's price sits within it.

**Valuation multiples**

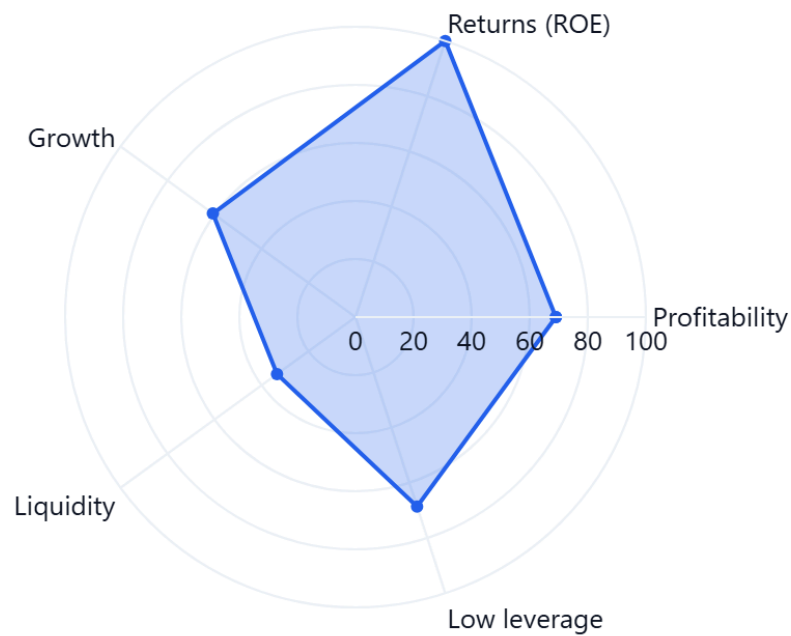


**What it shows.** The standard valuation lenses side by side — P/E, forward P/E, PEG, price-to-book, price-to-sales and EV/EBITDA — as horizontal bars. Seeing them together tells a more complete story than any single multiple.

**Where it is used.** Judging how richly the company is priced and flagging multiples that look stretched relative to the rest.

**Financial-health profile**

### Financial-health profile (0–100)



**What it shows.** Five dimensions — profitability, returns, growth, liquidity and low leverage — each normalised to a 0 to 100 score so they share one scale. The shape of the polygon is a quick fingerprint of where the company is strong and where it is weak. The scores are deliberately heuristic; the value is in the balance the shape communicates, not a precise benchmark.

**Where it is used.** A fast, holistic health check before drilling into any one metric.

### 9.3 Ask the analyst

The analyst tab sends the question to the reasoning agent, which calls the tools it needs and returns a concise, grounded answer.



# Stock Earnings Agent

Snapshot **Ask the analyst** Ingest documents

## Ask the analyst

The agent uses live financial data, SEC filings and your ingested documents.

e.g. How did AAPL's margins trend and what did management highlight?



In one sentence, what sector is AAPL in and its market cap?



Apple Inc. (AAPL) is in the Technology sector (Consumer Electronics industry) with a market cap of approximately \$4.87 trillion.

**Bottom line:** AAPL is a mega-cap technology/consumer electronics company valued at roughly \$4.87T.

## 9.4 Ingest documents

The ingestion tab accepts PDF, text and markdown files, optionally tags them with a ticker, and adds them to the searchable library.



# Stock Earnings Agent

Snapshot Ask the analyst Ingest documents

## Grow the knowledge base

Upload earnings PDFs or transcripts to make them searchable by the agent.

Associate with company (optional)

— none —

Document type

earnings\_call

PDF / text / markdown files

📎 Upload

200MB per file • PDF, TXT, MD, MARKDOWN

Ingest

## 10. Verification and Results

The screenshot below shows the health check, the test suites, and the retrieval evaluation running successfully. The health check confirms live connectivity to the model, the embeddings API and the vector store. Retrieval scores a perfect hit rate and mean reciprocal rank on the seeded evaluation set.

```

$ python -m scripts.healthcheck
stock-earnings-agent health check
-----
[OK] configuration: model=qwen/qwen3.8-27b, embedding=gemini-embedding-001
[OK] google embeddings: 3072-dimensional vector
[OK] vector store: collection holds 1 chunk(s)
[OK] groq llm: responded ('ok')
-----
All systems operational.

$ python -m pytest
29 passed, 6 deselected in 25.01s

$ python -m pytest tests/test_vector_store.py -m integration
6 passed in 14.44s

$ python -m evaluation.ragas_eval
Retrieval metrics: {'samples': 3, 'hit_rate': 1.0, 'mrr': 1.0}

```

| Check                                                | Result          |
|------------------------------------------------------|-----------------|
| Health check (config, embeddings, vector store, LLM) | All operational |
| Unit tests                                           | 29 passed       |
| Integration tests                                    | 6 passed        |
| Retrieval evaluation (hit rate, MRR)                 | 1.0, 1.0        |
| Lint (Ruff)                                          | Clean           |

## 11. Testing and Quality

Testing is split into two suites for a practical reason. ChromaDB's native core conflicts with the data frame libraries used elsewhere when both run in one process on Windows. The vector store tests are therefore marked as integration tests and run in their own process, while the default test run covers the rest.

- **Unit tests** mock all network access, so they run offline and quickly. They cover configuration validation, chunking, the embedding client, the financial tools, and filing retrieval.
- **Integration tests** exercise the real vector store with deterministic offline embeddings.
- **Static analysis** with Ruff enforces a consistent style and catches common errors.

## 12. Deployment

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The application ships with a container image. Dependencies are installed in a cached layer, the application runs as a non root user, and a built in health check acts as a readiness probe.

```
docker build -t stock-earnings-agent .  
docker run --rm -p 8501:8501 --env-file .env stock-earnings-agent
```

The MCP server can be run from the same image or directly:

```
python -m mcp_server.server
```

Configuration is supplied entirely through environment variables, which suits container and cloud deployment.

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## 13. Limitations

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- **Inference tier limits.** On a free hosted inference tier, output tokens per minute are capped. Complex multi step questions may reach that cap. A paid tier and a higher token limit resolve this.
  - **Data source coverage.** Market data reflects what the upstream provider exposes and may lag or omit fields for some tickers.
  - **Not financial advice.** The system produces analysis for research support, not investment recommendations.
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## 14. Future Work

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- **Delivered:** charting and historical trend visualisation in the snapshot view (candlestick price, quarterly fundamentals and margins, analyst target range, valuation multiples, financial-health radar and KPI gauges — see section 9.2).
  - Support peer and sector comparison across multiple tickers (treemap and scatter views).
  - Add earnings-surprise (actual versus estimated EPS) and analyst rating-trend charts.
  - Add scheduled ingestion of new filings as they are published.
  - Extend the evaluation set and track metrics over time.
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## 15. References

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- Model Context Protocol: <https://modelcontextprotocol.io/>
- FastMCP: <https://github.com/jlowin/fastmcp>
- LangGraph: <https://langchain-ai.github.io/langgraph/>
- LangChain: <https://python.langchain.com/>
- Groq documentation: <https://console.groq.com/docs>
- Google Gemini embeddings: <https://ai.google.dev/gemini-api/docs/embeddings>
- ChromaDB: <https://docs.trychroma.com/>
- Streamlit: <https://docs.streamlit.io/>
- yfinance: <https://github.com/ranaroussi/yfinance>
- SEC EDGAR access: <https://www.sec.gov/os/accessing-edgar-data>
- pdfplumber: <https://github.com/jsvine/pdfplumber>
- RAGAS: <https://docs.ragas.io/>
- Pydantic Settings: [https://docs.pydantic.dev/latest/concepts/pydantic\\_settings/](https://docs.pydantic.dev/latest/concepts/pydantic_settings/)
- Loguru: <https://github.com/Delgan/loguru>
- Tenacity: <https://tenacity.readthedocs.io/>
- pytest: <https://docs.pytest.org/>
- Ruff: <https://docs.astral.sh/ruff/>